Selection Rules of the Vector-Photon Vertex on S^3: the Physical Vertex, the Vertex Model and the Graph

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Abstract

We compute the electron–photon vertex $V = \int_{S^3} \langle \psi_1, c(A)\, \psi_2 \rangle$ on the round unit three-sphere, with $\psi_{1,2}$ Camporesi–Higuchi Dirac eigenspinors, A a coclosed 1-form eigenform and c Clifford multiplication, and use it to score eight selection rules. Representation theory of $\mathrm{Spin}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ gives the rule, and two independent exact computations confirm it with no extra channel and no accidental zero: a group-theoretic route through Clebsch–Gordan coefficients (896 cells: sign pair, photon component and levels $n_1, n_2 \le 3$, $q \le 7$) and a coordinate route by exact monomial integration in $\mathbb{R}^4$ (360 cells), which agree wherever both run. For spinors of the same sign of D the vertex keeps the window $|n_1 - n_2| < q < n_1 + n_2$, $n_1 + n_2 + q$ odd, of the vertex model used for spectral sums in Paper 28, and adds the channel $q = n_1 + n_2 + 1$. For opposite signs it opens channels with $n_1 + n_2 + q$ even, which the model has none of. The ground state couples, also to itself: the diagonal vertex $V(a, a)$ does not vanish. The Ward–Takahashi identity $\langle u, c(df)\, v \rangle = (\lambda_u - \lambda_v)\langle u, f v \rangle$ holds exactly. Each of the eight rules is scored three ways: on the GeoVac graph (scalar Fock graph 1/8, native Dirac graph 4/8, the latter by the definition of its edges), in the vertex model, and with the physical vertex. With the physical vertex, rules 1 (triangle sparsity) and 4 (Δm) hold, rule 6 holds in Ward–Takahashi form, and rules 2 (ground-state zero), 5 (E1 parity) and 8 ($q \le n_1 + n_2$) fail; rule 3 holds as a statement of representation theory but not as the model's channel-count table; for rule 7 the diagonal-vertex zero fails and Furry's theorem itself is not decided without charge conjugation and three-vertex traces. The 1+6+1 partition that the vertex model gives does not survive the physical vertex. Kept from the graph and model analyses: the pendant-edge value $\Sigma_{\mathrm{graph}}(\mathrm{GS}) = 2(n_{\max} - 1)/n_{\max}$, the fact that the edge Laplacian does not block-diagonalize by photon labels (q, m_q), the diagonal α_z identity of the model, and the normalization identity $\Sigma_{l=1} = 1/(4\pi)$. Coupling normalization, regularization and the gauge sector are owed.

1 Introduction

Paper 28 [6] studies the periods of spectral sums over the Camporesi–Higuchi Dirac spectrum [1] on S^3. Its nested sums couple spinor levels n_1, n_2 through a photon level q by a vertex model: the triangle rule for scalar harmonics, $|n_1 - n_2| \le q \le n_1 + n_2$, with $n_1 + n_2 + q$ odd imposed. This paper asks what the vertex of QED on round S^3 actually allows, and scores a list of eight selection rules against it.

The eight rules are those of Table 1: five standard rules of perturbative QED and three stated in terms of the S^3 construction. Each is scored three ways:

1. *on the graph*: QED built on the finite GeoVac Fock graph, with photons as 1-cochains (Sec. 4);
2. *in the vertex model*: vector-photon labels (q, m_q) with Wigner-3j coupling to scalar harmonics and the imposed parity rule, the model behind Paper 28's sums (Sec. 3);

3. *with the physical vertex*: the Clifford-multiplication vertex between Dirac eigenspinors and coclosed 1-forms, computed exactly (Sec. 2).

The physical vertex is the reference. It is computed at the level of selection rules and unnormalized amplitudes; the coupling normalization, the photon propagator with its gauge sector, and the flat-space limit are owed (Sec. 6).

#	Rule	Statement
1	Triangle sparsity	Clebsch–Gordan triangle conditions; for the spinor vertex on S^3, in both SU(2) fa
2	Vertex parity / ground-state zero	$n_1 + n_2 + q$ odd; hence the ground state does not couple
3	SO(4) channel count	the number W of open photon components vanishes on disallowed triples
4	Δm conservation	magnetic quantum numbers add at the vertex
5	Spatial parity (E1)	photon parity -1; same-parity states decouple
6	Ward identity	gauge modes decouple ($k_\mu \mathcal{M}^\mu = 0$)
7	Charge conjugation (Furry)	closed fermion loops with an odd number of vertices vanish
8	Level triangle	$\lvert n_1 - n_2\rvert \le q \le n_1 + n_2$

Table 1: The eight rules. Rules 1, 4, 5, 6, 7 are standard rules of perturbative QED; rules 2, 3 and 8 are stated for the S^3 construction. Levels are Camporesi–Higuchi levels $n \ge 0$ throughout; in the Fock labels $n_F = n + 1$ of the graph, rule 8 reads $\lvert n_{1F} - n_{2F}\rvert \le q \le n_{1F} + n_{2F} - 2$.

2 The Physical Vertex on S^3

2.1 Content

Write $S^3 = \mathrm{SU}(2)$, so that $\mathrm{Spin}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ acts by isometries. The Dirac operator has eigenvalues $\pm(n + 3/2)$ with $(n + 1)(n + 2)$ states for each sign [1]; with the convention used throughout, the sign-$+$ eigenspace at level n carries $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ content $((n+1)/2, n/2)$ and the sign-$-$ eigenspace $(n/2, (n+1)/2)$. The coclosed 1-forms at level $q \ge 1$ have Hodge eigenvalue $(q + 1)^2$ and two components,

$$A : \left(\tfrac{q+1}{2}, \tfrac{q-1}{2}\right), \qquad B : \left(\tfrac{q-1}{2}, \tfrac{q+1}{2}\right), \tag{1}$$

with $q(q+2)$ states each [2], on which $*d = \mp(q+1)$. Clifford multiplication by a 1-form is Spin(4)-equivariant, so V can be nonzero only if $\rho_1^* \otimes \rho_A \otimes \rho_2$ contains the trivial representation. SU(2) tensor products are multiplicity-free, so each (sign pair, component) carries at most one reduced amplitude.

2.2 The rule

Theorem 1 (Selection rule of the physical vertex)**.** *Let $d = \lvert n_1 - n_2\rvert$. The vertex can be nonzero only on the following channels.*

1. Same sign *($++$; $--$ is the mirror $A \leftrightarrow B$): $n_1 + n_2 + q$ odd, and component A for $d + 1 \le q \le n_1 + n_2 + 1$, component B for $d + 1 \le q \le n_1 + n_2 - 1$.*

2. Opposite sign *($+-$; $-+$ is the transpose): n_1+n_2+q even, with component A for $\lvert n_1+1-n_2\rvert \le q + 1 \le n_1 + n_2 + 1$ and $\lvert n_1 - n_2 - 1\rvert \le q - 1 \le n_1 + n_2 + 1$, and component B for $\lvert n_1 + 1 - n_2\rvert \le q - 1 \le n_1 + n_2 + 1$ and $\lvert n_1 - n_2 - 1\rvert \le q + 1 \le n_1 + n_2 + 1$.*

Proof. The triangle and integrality conditions in each SU(2) factor applied to the content above. For same-sign spinors the left triangle $((n_1+1)/2, (q\pm1)/2, (n_2+1)/2)$ and the right triangle $(n_1/2, (q\mp1)/2, n_2/2)$ give the stated ranges, and all four integrality conditions reduce to n_1+n_2+q odd. For opposite signs the content of ψ_2 is mirrored, which shifts each triangle by one unit and flips the integrality condition to n_1+n_2+q even. □

Corollary 1 (Comparison with the vertex model)**.** *With n_1+n_2+q odd, the model window $|n_1-n_2| \le q \le n_1+n_2$ is $d+1 \le q \le n_1+n_2-1$. On it both same-sign components are open, so the physical same-sign count equals the model's $W=2$ there (Paper 28, Sec. "Inner photon sum"). The only same-sign channel outside it is $q=n_1+n_2+1$, with one component. Every opposite-sign channel is outside the model.*

Corollary 2 (The ground state couples)**.** *For an external level $n=0$ the open channels are: same sign at $q=n_{\rm int}+1$ for every $n_{\rm int}\ge 0$, and opposite sign at $q=n_{\rm int}$ for every $n_{\rm int}\ge 1$ (component B for external sign +), one component each. In particular the ground state couples to itself at $q=1$.*

The theorem is a necessary condition. That each open channel has a nonzero reduced amplitude is checked below.

2.3 Exact computation by two routes

The rule and the amplitudes were computed in exact arithmetic before comparison with the predictions, which were registered first (prediction record `docs/predictions/2026-11-07_physical_vector_vertex.md`, with its scored outcome); both routes are ported into the regression suite as `tests/test_paper33_physical_vertex`

- *Route 1 (group theory).* Functions on SU(2) are Wigner D-functions; spinors and 1-forms are built in the left-invariant frame, $D = 2\,\sigma\cdot R + 3/2$ and $*d = -2\,S\cdot R - 2$, with R the right-translation generators and the constants fixed by the spin connection and by $*d(\text{grad}) = 0$. The vertex factorizes into a left Clebsch–Gordan coefficient and a frame-side contraction; zero is exact zero in the multi-quadratic number field. Range $n_1, n_2 \le 3$, $q \le 7$, both signs, both components, all m: 896 cells.

- *Route 2 (coordinates in $\mathbb{R}^4$).* Eigenspinors are exact null spaces of $D-\lambda$ on polynomial spinors restricted to S^3, checked against D directly; eigenforms are exact null spaces of $*d \mp (q+1)$ on polynomial 4-vectors, with tangency and coclosedness verified afterwards, not imposed. Content is measured by the two su(2) Casimirs. Vertices are integrated by exact monomial moments on S^3, with no use of representation theory. Range $n_1, n_2 \le 2$, $q \le 5$: 360 cells. Route 2 imports neither route 1 nor the `geovac` package.

Controls: both routes reproduce the Dirac spectrum $\pm(n+3/2)$ with multiplicity $(n+1)(n+2)$ per sign and the coclosed spectrum with $q(q+2)$ states per component; route 1 restricted to the model window reproduces the channel counts of `geovac/qed_anomalous_moment.py`; and the two routes agree on all 360 common cells, on open/closed and on the exact value of $\sum_m |V|^2$. On all 896 cells $\sum_m |V|^2$ equals the closed form

$$\sum_m |V|^2 = 6\,(2j_A{+}1)(2j_2{+}1)(2j_1{+}1)(2J_A{+}1)(2J_2{+}1)(2J_1{+}1) \begin{Bmatrix} j_A & j_2 & j_1 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ J_A & J_2 & J_1 \end{Bmatrix}^2 , \qquad (2)$$

where for each field j is the left-translation spin of its scalar part and J the right-translation spin including the frame index ($J = j \pm 1/2$ for a spinor, $J_A = j_A \pm 1$ for a coclosed 1-form).

Results. Of the 896 cells, 168 are open (44 ++, 44 −−, 40 +−, 40 −+); every open cell obeys Theorem 1 and every cell the theorem allows is open. Of the 88 open same-sign cells, 32 lie outside the model window, all at $q = n_1 + n_2 + 1$. Every nonzero amplitude conserves m in each SU(2) factor separately. Computed with Haar measure of volume one, every amplitude is algebraic; π enters only with the volume $2\pi^2$ of the round sphere.

2.4 The diagonal vertex

Proposition 1 (The diagonal vertex does not vanish)**.** *For each ground state a ($n = 0$, either sign of D), $\sum_{m_A} |\langle a|c(A)|a\rangle|^2 = 1$ over the $q = 1$ coclosed eigenforms of the open component (A for sign +, B for sign −; the other component gives 0).*

Both routes give 1 for every basis state. The witness is elementary: a Killing spinor (constant in the left-invariant frame) against a left-invariant coframe 1-form, a coclosed $q = 1$ eigenform, gives a Pauli matrix element times the volume.

2.5 Ward–Takahashi identity

Proposition 2 (Ward–Takahashi identity on S^3)**.** *For Dirac eigenspinors u, v with eigenvalues λ_u, λ_v and a function f,*

$$\langle u, c(df)\, v\rangle = (\lambda_u - \lambda_v)\, \langle u, fv\rangle. \tag{3}$$

Proof. $[D, f] = c(df)$ as operators; take matrix elements between eigenspinors. □

Route 2 checks Eq. (3) exactly on 384 element pairs (f a harmonic polynomial of degree 1 or 2, spinor levels $n \le 1$), 56 of them with nonzero left side. The identity is the vertex-level form of the Ward–Takahashi identity [3, 4]: an exact (pure-gauge) photon decouples between degenerate states, and in general its coupling is fixed by the charge vertex.

2.6 An unnormalized one-loop sum

With the physical channels and both internal signs, the one-loop sum of Paper 28's self-energy form at external level 0, with the squared amplitudes in place of the channel count (unit coupling, volume-one measure, photon weight $q(q+2)$ as in Paper 28), is 0.2669, 0.2702, 0.2715, 0.2719 at cutoffs $n_{\max} = 3, 5, 8, 12$. It is nonzero at every cutoff, where the model's corresponding sum is zero (Paper 28, Theorem 4). The number is not a self-energy: normalization, regularization and the photon's Hodge eigenvalue $(q+1)^2$ in place of $q(q+2)$ are owed.

3 The Vertex Model

3.1 Construction

Definition 1 (Vertex model)**.** *Photon modes are (q, m_q), $1 \le q \le q_{\max}$, $|m_q| \le q$, with propagator $G_\gamma(q) = 1/[q(q+2)]$. For electron states $a = (n_a, l_a, m_a)$, $b = (n_b, l_b, m_b)$ the vertex is*

$$V(a, b, q, m_q) = \begin{pmatrix} l_a & q & l_b \\ -m_a & m_q & m_b \end{pmatrix} \theta(l_a + l_b + q \text{ odd}), \tag{4}$$

times the harmonic normalization $\sqrt{(2l_a+1)(2q+1)(2l_b+1)/(4\pi)}$ and unit radial factor; for Dirac electrons the photon couples through α_z against the scalar harmonic $Y_q^{m_q}$. It is implemented in `geovac/vector_qed.py`.

This is the vertex model of Paper 28 in angular form: its triangle is the scalar-harmonic one and its parity rule is imposed, so its ground-state self-energy vanishes for the reason of Paper 28, Theorem 4.

3.2 The diagonal α_z identity

Theorem 2 (Diagonal α_z vertex vanishes)**.** *For Dirac spinors $\psi = (f\Omega_\kappa, ig\Omega_{-\kappa})$ with real radial functions and the model's coupling, α_z against $Y_q^{m_q}$, the diagonal vertex satisfies $V(a,a,q,m_q) = 0$ identically.*

Proof. For $a = b$ the m-rule forces $m_q = 0$, so only Y_q^0 and $\alpha_z = \left(\begin{smallmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{smallmatrix}\right)$ enter. Then

$$\psi^\dagger \alpha_z \psi = ifg\left[\Omega_\kappa^\dagger \sigma_z \Omega_{-\kappa} - \Omega_{-\kappa}^\dagger \sigma_z \Omega_\kappa\right] = -2fg\,\Im M, \qquad M = \Omega_\kappa^\dagger \sigma_z \Omega_{-\kappa}.$$

Expanding $\Omega_{j,l,m_j} = \sum_{m_s} \langle l, m_l; \tfrac{1}{2}, m_s | j, m_j\rangle Y_l^{m_l} \chi_{m_s}$ with $m_l = m_j - m_s$, σ_z is real diagonal and conserves m_s, so

$$M = \sum_{m_s = \pm 1/2} (\operatorname{sign} m_s) \langle l_\kappa, m_l; \tfrac{1}{2}, m_s | j, m_j\rangle \langle l_{-\kappa}, m_l; \tfrac{1}{2}, m_s | j, m_j\rangle Y_{l_\kappa}^{m_l\,*} Y_{l_{-\kappa}}^{m_l},$$

with real Clebsch–Gordan coefficients. The product $Y_{l_\kappa}^{m_l\,*} Y_{l_{-\kappa}}^{m_l}$ has matched magnetic indices, so its phases $e^{\pm i m_l \phi}$ cancel and it is a real function of θ. Hence $\Im M \equiv 0$, the integrand vanishes pointwise, and $V(a,a,q,0) = \int \psi^\dagger \alpha_z \psi\, Y_q^0\, d\Omega = 0$. □

Six cases ($\kappa \in \{-1, +1, -2\}$, $m_j \in \{\pm 1/2, +3/2\}$) are verified symbolically in `tests/test_paper33_furry_deri` the bilinear vanishes pointwise, $\Im M \equiv 0$ with $M \not\equiv 0$, and the projected integral is zero. The mechanism is m_l conservation, not Hermiticity alone: Hermiticity gives $V_{LS} + V_{SL} = 2i\,\Im V_{LS}$, and $\Im V_{LS} = 0$ needs the matched magnetic index.

Remark 1 (Not a property of the physical vertex)**.** *Theorem 2 concerns the model's coupling. A physical vector photon with $m_q = 0$ also has $m = \pm 1$ spherical components, which couple through $\alpha_\pm$; the diagonal magnetic coupling is nonzero, as Proposition 1 shows on S^3. Nor would a diagonal zero be Furry's theorem [5]: in a closed loop the intermediate vertices connect different states, so a diagonal identity removes only the tadpole, while Furry's theorem comes from charge conjugation of the field. In `geovac/vector_qed.py` the diagonal zero is imposed by setting $V(a,a) = 0$.*

3.3 The normalization identity

Proposition 3 ($\Sigma_{l=1} = 1/(4\pi)$)**.** *In the vertex model at $n_{\max} = 2$ the $l = 1$ self-energy is exactly $1/(4\pi)$.*

Proof. Only $q = 1$ and the two $2p$ intermediate states b contribute, with $4\pi \sum_{m_q} |V|^2 = 9/2$ per b (the Gaunt integral with the harmonic normalization of Definition 1), $G_e = 1/3$ and $G_\gamma(1) = 1/3$, so $\Sigma_{l=1} = 2 \cdot \frac{9}{2} \cdot \frac{1}{9} \cdot \frac{1}{4\pi}$. Verified in `test_scalar_l1_self_energy_is_one_over_4pi` (`tests/test_vector_qed.py`). □

The factor $1/(4\pi)$ is the normalization of spherical harmonics on the unit S^2, which the model's angular vertex carries. The physical vertex of Sec. 2, normalized to volume one on S^3, does not carry it.

4 QED on the Graph

4.1 Construction

On the finite Fock graph (nodes (n, l, m_l), edges the Fock ladders) photons are 1-cochains, the photon propagator is the pseudo-inverse $G_\gamma = L_1^+$ of the edge Laplacian $L_1 = B^\top B$ (B the incidence matrix), and the vertex places the two electron legs at the endpoints of the photon edge. The native Dirac graph replaces the node labels by (n, κ, m_j), with edge rule B (E1 dipole: $\Delta l = \pm 1$, $|\Delta j| \leq 1$, $|\Delta m_j| \leq 1$).

4.2 The pendant-edge value

Proposition 4 (Pendant-edge theorem)**.** *On the scalar Fock graph at every $n_{\max} \geq 2$, the one-loop self-energy at the ground state $|1, 0, 0\rangle$ is*

$$\Sigma_{\text{graph}}(\text{GS}) = \frac{2(n_{\max} - 1)}{n_{\max}} \xrightarrow{n_{\max} \to \infty} 2.$$

Proof. The ground state is a leaf of the Fock graph: its unique edge e_0 joins it to $|2, 0, 0\rangle$. The s-wave edges form a path of $N = n_{\max} - 1$ edges in L_1, decoupled from the $l \geq 1$ edges, and the path-graph Laplacian satisfies $T_N^{-1}[0, 0] = N/(N + 1)$, so $G_\gamma[e_0, e_0] = (n_{\max} - 1)/n_{\max}$. All Clebsch–Gordan coefficients at $l = 0$ equal 1 and the two intermediate states $m_j = \pm 1/2$ contribute equally, giving $\Sigma = 2\, G_\gamma[e_0, e_0]$. □

Verified in `tests/test_graph_qed_self_energy.py`: exactly at $n_{\max} = 2$ (ground-state block $[[1, 1], [1, 1]]$) and numerically to 10^{-10} at $n_{\max} = 3, 4$. The graph value is nonzero, like the physical vertex's and unlike the model's.

4.3 Photon labels are not in the edge Laplacian

Observation 1 (L_1 does not block-diagonalize by (q, m_q))**.** *Photon labels describe the transition between the endpoint labels of an edge, not a quantity conserved at a vertex: at a vertex, edges with different photon labels meet, and L_1 couples them. The cross-sector Frobenius fraction of L_1 in the (q, m_q) grouping is 20% at $n_{\max} = 3$ on the scalar Fock graph and 57% on the Dirac rule-B graph. The only decompositions are by connected components (l on the Fock graph, κ on the Dirac rule-A graph).*

So vector-photon structure is not a reorganization of the graph; it has to be added.

4.4 Graph scores

On the scalar Fock graph only rule 1 holds: the Fock edges obey the ladder selection rules, which are Clebsch–Gordan triangle conditions. On the Dirac rule-B graph rules 1, 4, 5 and a graph analog of rule 7 hold, all by the definition of the edges: m_j is a node label, $\Delta l = \pm 1$ is the edge rule, and the bipartite structure by l-parity makes the identity-vertex tadpole vanish. These are construction-consistency checks, not computed recoveries. Rule 8 fails on both graphs because the edges are nearest-neighbour in n. Rule 6 was scored on the graph by a commutator proxy, $\|[\Sigma, H_0]\|/\|\Sigma\|$, which is not a Ward identity; it fails there. The aggregate scores 1/8 and 4/8 are asserted in `tests/test_vector_qed.py` (`TestCensusAggregates`).

5 The Three-Way Census

Rule	Scalar Fock graph	Dirac graph (B)	Vertex model	Physical vertex
1. Triangle sparsity	holds	holds	holds	holds (both factors)
2. Parity / GS zero	fails	fails	holds	fails
3. Channel count	fails	fails	holds[a]	holds[b]
4. Δm	fails	holds[c]	holds	holds (both factors)
5. Spatial parity (E1)	fails	holds[c]	holds[d]	fails
6. Ward identity	fails[e]	fails[e]	—[e]	holds (WT form)
7. Furry	fails	holds[c,f]	holds[g]	undecided[h]
8. Level triangle	fails	fails	holds	fails

Table 2: Each rule scored on the graph, in the vertex model, and with the physical vertex. [a] By construction: $W = 2$ on the model window and 0 off it. [b] As representation theory (no extra channel, no accidental zero); the model's W table does not hold, since $W_{\text{phys}} = 1$ where $W_{\text{model}} = 0$ at every $q = n_1 + n_2 + 1$. [c] By the definition of the rule-B edges. [d] Imposed as the parity step of Eq. (4). [e] Scored by the commutator proxy $\|[\Sigma, H_0]\|/\|\Sigma\|$, which tests block-diagonality in n, not gauge invariance; in the model it passes at $n_{\max} = 2$ and fails at $n_{\max} = 3$ (ratio 0.61 Dirac, 3.1 scalar electrons), and the model has no gauge sector in which a Ward identity can be stated. [f] Graph analog: the identity-vertex tadpole vanishes. [g] Dirac electrons only, through Theorem 2; for scalar electrons $V(a, a, q, m_q) \neq 0$ at odd q and the rule fails. [h] The diagonal-vertex mechanism fails (Proposition 1); Furry's theorem itself needs charge conjugation on S^3 spinors and three-vertex traces.

Rule by rule, with the physical vertex.

(1) Holds in its Spin(4) form: every open cell satisfies both SU(2) triangles.

(2) Fails: the ground state couples (Corollary 2), and opposite-sign channels have $n_1 + n_2 + q$ even. The parity clause is a rule for same-sign spinors only.

(3) Holds as the statement that the vertex vanishes on representation-theoretically disallowed cells; fails as the model's table, which misses $q = n_1 + n_2 + 1$ and every opposite-sign channel.

(4) Holds, in each SU(2) factor separately.

(5) Fails in its literal form, since a state couples to itself (Proposition 1). The antipodal element of Spin(4) gives photon parity $(-1)^{q+1}$, not -1; the selection content it carries is the integrality condition already in rules 1 and 2.

(6) Holds as the Ward–Takahashi identity, Eq. (3).

(7) The diagonal-vertex zero fails; Furry's theorem is not decided here.

(8) Fails at every $q = n_1 + n_2 + 1$.

What the model's tally was, and why it does not carry over. In the vertex model the rules grouped as 1+6+1: rule 1 from the graph, rules 2–6 and 8 from adding the photon labels, and rule 7 from the α_z identity (Dirac electrons). With the physical vertex that partition does not survive: three of the six rules the photon labels were credited with fail (2, 5, 8), the channel

count holds only as representation theory, rule 6 holds in a form the model cannot state, and the mechanism behind rule 7 fails. What does carry over is that rules 1 and 4 are properties of the coupling coefficients in every construction.

6 Open Questions

1. *Normalization.* Fix the coupling and the photon field normalization: the transverse propagator on S^3 in a stated gauge, with the coexact mode measure and Hodge eigenvalue $(q+1)^2$, and the relation of the volume-one amplitudes to the radius.
2. *Regularization and the flat-space limit.* The large-level asymptotics of the reduced amplitudes, from Eq. (2), are needed to define the one-loop sums and to take $R \to \infty$ at fixed physical momentum.
3. *Gauge sector.* Include the exact and longitudinal modes and the ghosts (Feynman gauge), with Eq. (3) as input.
4. *Furry's theorem.* Construct charge conjugation on S^3 spinors and compute three-vertex traces to decide rule 7.
5. *Periods.* Which periods do the chain sums of Paper 28 take when built on the physical window and amplitudes?

7 Conclusion

The vertex of QED on round S^3, computed exactly by two independent routes, keeps the vertex model's rule for same-sign spinors inside the model's window and goes beyond it in three ways: the channel $q = n_1+n_2+1$, opposite-sign channels with n_1+n_2+q even, and a nonzero diagonal vertex, so that the ground state couples. It satisfies the Ward–Takahashi identity exactly. Scored against it, two of the eight rules hold as stated (triangle sparsity, Δm), one holds in Ward–Takahashi form, one holds only as representation theory, three fail, and Furry's theorem is left undecided. The graph scores (1/8 and 4/8) and the model's 1+6+1 tally describe those constructions, not the physical vertex.

The stage-A computation is recorded in the prediction record named in Sec. 2; a regression test porting its checks to `tests/` is owed.

References

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